1.1 T/F Questions

1-5: TFTFF 6-10: FTFFF 11-15: FFTTF

1.3 Standard Questions

1.3.1

If x and g are elements of a group G, prove that $|x| = |gxg^{-1}|$.
Deduce that $|ab| = |ba|$ for all $a, b \in G$.
Proof: Note that $(gxg^{-1})^n = gx^ng^{-1}$ so $x^n = e \iff (gxg^{-1})^n = e$ and $|x| = |gxg^{-1}|$. $ba = b(ab)b^{-1}$ so $|ab| = |ba|$.

1.3.2

Let $n, m \in \mathbb{N}_{>2}$. Determine all homomorphisms $\phi : \mathbb{Z}_n \longrightarrow D_{2m}$.
Solution: $\mathbb{Z}_n = \langle 1 \rangle$ so $\mathrm{Hom}(\mathbb{Z}_n, D_{2m})$ correspond to elements of D_{2m} with order a factor of n. Suppose $D_{2m} = \langle a, b \mid a^m = b^2 = e, ab = ba^{-1} \rangle$, then $|a^k| = m/\gcd(m, k)$ and $|ba^k| = 2$.
Hence if n is even, then the homomorphisms are $1 \mapsto ba^k$ and $1 \mapsto a^k, m \mid nk$.
Otherwise they are $1 \mapsto a^k, m \mid nk$.

1.3.3

Prove that S_n is generated by (12) and $(12 \cdots n)$. One can probably prove this directly, but one can also prove it as listed in the class, by first showing S_n is generated by all (ij), then showing S_n is generated by $(i, i+1)$, and finally proving this statement.
Proof: First every $\sigma \in S_n$ can be decomposed into cycles, which are generated by all (ij). Then note that $(i, i+k) = (i, i+1)(i+1, i+k)(i, i+1)$ so by induction (ij) are generated by all $(i, i+1)$. Note that $(i, i+1) = (12 \cdots n)^{-1}(i-1, i)(12 \cdots n)$ so S_n is generated by (12) and $(12 \cdots n)$.

1.3.4

Show that an element has order 2 in S_n if and only if its cycle decomposition is a product of commuting 2-cycles. In S_6, how many elements have order 2?
Proof: If $\sigma \in S_n$ has order 2, then $\sigma(\sigma(i)) = i \forall i$, so we can pair i with $\sigma(i)$ to obtain a product of disjoint (commuting) 2-cycles whose product is σ. The converse is trivial.
For S_6: one 2-cycle: $C_6^2 = 15$; two 2-cycles: $C_6^2 C_4^2/2 = 45$; three 2-cycles: $C_6^2 C_4^2/3! = 15$. Hence a total of 75 cycles.

1.3.5

Prove that D_8 and Q_8 are not isomorphic. Find the definition in Dummit-Foote.
Proof: Recall $D_8 = \langle a, b \mid a^4 = b^2 = 1, ab = ba^{-1} \rangle$ and $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, so D_8 has 6 elements of order 2, while Q_8 has only 1. Hence they are not isomorphic.

1.3.6

Prove that every subgroup of a cyclic group is cyclic, and every quotient group of a cyclic group is cyclic.
Proof: Suppose $H < G = \langle a \rangle$, take the least k such that $a^k \in H$, then clearly $H = \langle a^k \rangle$. Clearly $G/H = \mathbb{Z}/k\mathbb{Z}$ is also cyclic.

1.3.7

Let G be a group. Define an opposite group G^{op} with law of composition $a \star b$ as follows: the underlying set is the same as G, but the law of composition is $a \star b = ba$.

- Prove that G^{op} is a group.
- Prove that G and G^{op} are isomorphic.
  Proof: The identity is e, the inverse of a is a^{-1}, and $(a \star b) \star c = cba = a \star (b \star c)$. There is a natural isomorphism $G \to G^{op}$, $a \mapsto a^{-1}$.

1.3.8

Let $\phi : G \to H$ be a group homomorphism, and let K be a subgroup of H.

1. Show that $\phi^{-1}(K)$ is a subgroup of G.
2. When E is a normal subgroup of H, show that $\phi^{-1}(E)$ is normal in G.
3. When G is finite, show that

$$|\phi^{-1}(K)| = |\ker(\phi)| \cdot |\phi(G) \cap K|.$$

   Note: one needs to first show the finiteness of each term.
   Proof: (1) If $a, b \in \phi^{-1}(K)$ then $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} \in K$ so $ab^{-1} \in \phi^{-1}(K)$.
   (2) If $a \in \phi^{-1}(K)$ then $\phi(gag^{-1}) = \phi(g)\phi(a)\phi(g)^{-1} \in \phi^{-1}(K)$.
   (3) Clearly $\phi^{-1}(K), \ker(\phi) \subset G$ is finite, and $\phi(G) \cap K \subset \phi(G)$ is finite.
   Let $\bar{\phi} : G/\mathrm{Ker}\phi \to H$ then $\bar{\phi}$ is injective and $\bar{\phi}(\bar{\phi}^{-1}(K)) = \mathrm{Im}\bar{\phi} \cap K$ so $|\bar{\phi}^{-1}(K)| = |\phi(G) \cap K|$.
   Note that $\bar{\phi}^{-1}(K)$ is the quotient $\phi^{-1}(K)/\mathrm{Ker}\phi$, so $|\phi^{-1}(K)| = |\mathrm{Ker}\phi| \cdot |\phi(G) \cap K|$.

1.3.9

Consider the group $G = \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ given in terms of the following generators and relations:

$$G = \langle x, y \mid x^4 = y^4 = 1,\ xy = yx \rangle.$$

Let $\overline{G} = G/\langle x^2y^2 \rangle$ (note that every subgroup of the abelian group G is normal).

1. Show that the order of $\overline{G}$ is 8.
2. Exhibit each element of $\overline{G}$ in the form $\overline{x}^{\,a}\overline{y}^{\,b}$ for some integers a and b.
3. Find the order of each of the elements of $\overline{G}$ exhibited in (2).
4. Give an explicit isomorphism $\overline{G} \simeq \mathbb{Z}_4 \times \mathbb{Z}_2$.
   Proof: (1) Since x^2y^2 has order 2, $|\bar{G}| = |G|/|\langle x^2y^2 \rangle| = 8$.
   (2) $\bar{G} = \{1, \bar{x}, \bar{y}, \bar{x}^2, \bar{x}\bar{y}, \bar{x}^3, \bar{x}^2\bar{y}, \bar{x}^3\bar{y}\}$.

(3) Their orders are respectively, 1,4,4,2,2,4,4,2.
(4) Consider the isomorphism $\bar{G} \to \mathbb{Z}_4 \times \mathbb{Z}_2, \bar{x} \mapsto (1,0), \bar{y} \mapsto (-1,1)$.

1.3.10

Let N be a finite subgroup of a group G. Show that, for an element $g \in G$, $gNg^{-1} \subseteq N$ if and only if $gNg^{-1} = N$. Can you give an example where this fails for an infinite N? Hint: you can consider a case similar to Problem 1.2.10.
Proof: Trivial since $|gNg^{-1}| = |N|$ for finite sets.
For infinite N, consider $G = \{f : \mathbb{R} \to \mathbb{R} \text{ bijective}\}$ and translation $T : x \mapsto x + 1$, dilation $D : x \mapsto 2x$, then $N = \langle T \rangle$ satisfy $DND^{-1} = \langle T^2 \rangle \subsetneq N$.

1.3.11

Let

$$D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$$

be the usual presentation of the dihedral group of order $2n$, and let k be a positive integer dividing n.

1. Prove that $\langle r^k \rangle$ is a normal subgroup of D_{2n}.
2. Prove that $D_{2n}/\langle r^k \rangle \cong D_{2k}$.
   Proof: Consider $\varphi : D_{2n} \to D_{2k} = \langle a, b \mid a^k = b^2 = 1, ab = ba^{-1} \rangle, r \mapsto a, s \mapsto b$ which is a well-defined homomorphism, then $\langle r^k \rangle = \mathrm{Ker}\phi$ is a normal subgroup $D_{2k} = \mathrm{Im}\varphi \cong D_{2n}/\mathrm{Ker}\phi = D_{2n}/\langle r^k \rangle$.

1.3.12

Any subgroup H of G of index 2 is normal.
Proof: For any $g \in G \setminus H$, $G = H \sqcup gH = H \sqcup Hg$, so $gH = Hg$ and hence $H \triangleleft G$.

1.3.13

Let H_1, H_2 be subgroups of a group G, and let N be a normal subgroup of G contained in $H_1 \cap H_2$. Show that

$$(H_1/N) \cap (H_2/N) \cong (H_1 \cap H_2)/N.$$

Here the intersection takes place in G/N.
Proof: Clearly they are the same subset of G/N.

1.3.14

Let G be a finite group. Let $H, K < G$ be subgroups and let $a \in G$. The set

$$HaK := \{hak \mid h \in H,\ k \in K\}$$

is called a double coset for H and K. Prove that

$$|HaK| = |H| \cdot [K : a^{-1}Ha \cap K].$$

Proof: Suppose $K = \bigsqcup_i (a^{-1}Ha \cap K)k_i$, then $hak = h'ak'$ implies $k(k')^{-1} = a^{-1}(hh')a \in a^{-1}Ha \cap K$, so Hak_i are disjoint sets. For each $hak \in HaK$ suppose $k \in (a^{-1}Ha \cap K)k_i$, then $kk_i^{-1} = a^{-1}h'a$ so $HaK = \bigsqcup_i Hak_i$.

1.3.15

Let H, K, N be subgroups of a group G such that

$$H < K, \qquad H \cap N = K \cap N, \qquad HN = KN.$$

Show that $H = K$.
Proof: Take $k \in K$. Since $KN = HN$, write $k = hn$ with $h \in H, n \in N$, then $n = h^{-1}k \in K \cap N = H \cap N \subset H$. Hence $k = hn \in H$ so $K = H$.

1.3.16

Suppose that H and K are subgroups of finite index in the (possibly infinite) group G, with $[G : H] = m$ and $[G : K] = n$. Prove that $\operatorname{lcm}(m, n) \leq [G : H \cap K] \leq mn$.
Deduce that if m and n are relatively prime, then

$$[G : H \cap K] = [G : H] \cdot [G : K].$$

Proof: $m \mid [G : H][H : H \cap K] = [G : H \cap K]$ and likewise $n \mid [G : H \cap K]$, so $\operatorname{lcm}(m, n) \mid [G : H \cap K]$.
Note that $[H : H \cap K] = [HK : K] \leqslant [G : K] = n$, so $[G : H \cap K] \leqslant mn$.

1.3.17

Suppose that H and K are both normal subgroups of G and that G/H and G/K are both solvable. Prove that $G/(H \cap K)$ is solvable.
Proof: Consider $\phi : G/(H \cap K) \to G/H \times G/K, g(H \cap K) \mapsto (gH, gK)$, so $G/(H \cap K)$ is a subgroup of $G/H \times G/K$ which is solvable, hence $G/(H \cap K)$ is solvable.

1.3.18

1. Let G be a finite abelian group with elements $a_1, a_2, \ldots, a_n$. (We use multiplicative convention in this problem.) Prove that $a_1 \cdots a_n$ is an element whose square is the identity.
2. If the G in part (1) has no elements of order 2 or more than one element of order 2, prove that $a_1 \cdots a_n = e$.
3. If G has exactly one element y of order 2, prove that $a_1 \cdots a_n = y$.
4. (Wilson's theorem) If p is a prime number, show that $(p-1)! \equiv -1 \pmod{p}$.
   Proof: (1) Since $g \mapsto g^{-1}$ is bijective (isomorphism), $(a_1 \cdots a_n)^{-1} = a_n^{-1} \cdots a_1^{-1} = a_1 \cdots a_n$.

(2) If there are no elements of order 2, clearly $a_1 \cdots a_n = e$. If more than one element has order 2, pair every element with its inverse, then only elements of order 2 contribute to the product. Take the normal subgroup $H = \mathrm{Ker}\phi$ with $\phi : g \mapsto g^2$, and take $u, v \in H$. Then the product of four elements of each coset of $H/\langle u, v\rangle$ is the identity, so $a_1 \cdots a_n = e$.
(3) Pair every element with its inverse, the product reduces to y.
(4) The only element of order 2 is clearly -1.

1.3.19

Let G be a group and let A be a subgroup and N a normal subgroup. Prove that AN is solvable if and only if both A and N are solvable.
Proof: A, N are subgroups of AN so one side is trivial.
If both A, N are solvable, then $AN/N \cong A/H \cap A$ is solvable, so AN is solvable.

1.4. More Challenging Questions

1.4.1

List all subgroups of the dihedral group D_{2n}. (Note: one has to separate the discussion according to whether n is even or odd.) Determine which subgroups are normal.
Proof: Suppose $H < D_{2n} = \langle a, b \mid a^n = b^2 = 1, ab = ba^{-1}\rangle$.
Case1: $H < \langle a\rangle$, then $H = \langle a^k\rangle$ with $k \mid n$, and they are all normal.
Case2: $H \not\subset \langle a\rangle$, suppose $\langle a^k\rangle = H \cap \langle a\rangle$ and take any $a^l b \in H$, then $\langle a^k, a^l b\rangle < H$. For any $a^t b \in H$, a^{t-l} $= a^t b \cdot a^l b \in H$ so $k \mid t - l$. Hence $a^t b \in \langle a^k, a^l b\rangle$ so $H = \langle a^k, a^l b\rangle$ where $k \mid n$ and $1 \leqslant k \leqslant n$. These are all subgroups. If n is odd, the only normal one is D_{2n}, and if $n = 2k$ is even, the normal ones are D_{2n} and D_{2k} $= \langle a^2, b\rangle$ of index 2.

1.4.2

Let H be a subgroup of a finite group G. Show that there exist elements $c_1, \dots, c_n$ simultaneously representing left and right cosets of H in G, that is, $G/H = \{c_1 H, \dots, c_n H\}$ and at the same time $H\backslash G$ $= \{Hc_1, \dots, Hc_n\}$.
Proof: Consider the bipartite graph $V_1 = G/H, V_2 = H \setminus G$, with an edge connecting gH, Hh iff $gH \cap Hh$ $\neq \emptyset$. Then clearly it satisfy the condition of Hall's theorem, so take a perfect matching, and let c_i be in the union of each pair.

1.4.3

Let G be the group of rigid motions in $\mathbb{R}^3$ of a cube (or an octahedron). Show that $|G| = 24$.
Give an explicit isomorphism $G \simeq S_4$.
Proof: A rigid motion is determined by the image of a face, and one edge on it, so $6 \cdot 4 = 24$ choices in total. Each rigid motion permutes the four diagonals, giving a homomorphism $\phi : G \to S_4$. The kernel is trivial, and they have the same order, so $G \cong S_4$.

1.4.4

1. Give a one-line proof that $\mathbb{Z}$ is a normal subgroup of $\mathbb{Q}$ (the additive subgroup).
2. Let p be a prime, and let $\mathbb{Z}(p^\infty)$ be the following subset of $\mathbb{Q}/\mathbb{Z}$:

$$\mathbb{Z}(p^\infty) = \left\{ \frac{a}{b} + \mathbb{Z} \in \mathbb{Q}/\mathbb{Z} \mid a, b \in \mathbb{Z},\ b = p^i \text{ for some } i \geq 0 \right\}.$$

Show that $\mathbb{Z}(p^\infty)$ is an infinite subgroup of $\mathbb{Q}/\mathbb{Z}$.

3. Show that every element of $\mathbb{Z}(p^\infty)$ has finite order p^n for some $n \geq 0$.
4. Show that the subgroup $H_n := \{z \in \mathbb{Z}(p^\infty) \mid p^n \cdot z = 0\}$ is isomorphic to $\mathbb{Z}/p^n\mathbb{Z}$.
5. Deduce that $\mathbb{Z}(p^\infty) = \bigcup_n H_n$.
6. Prove that $\mathbb{Z}(p^\infty)$ is divisible; that is, for each element $x \in \mathbb{Z}(p^\infty)$ and for each $n \in \mathbb{N}$, there exists $y \in \mathbb{Z}(p^\infty)$ such that $x = ny$.
   Proof: (1) $\mathbb{Q}$ is abelian so any subgroup is normal.
   (2) $\left(\frac{a}{p^i} + \mathbb{Z}\right) - \left(\frac{b}{p^j} + \mathbb{Z}\right) = \frac{ap^j - bp^i}{p^{i+j}} + \mathbb{Z}$ so $\mathbb{Z}(p^\infty) < \mathbb{Q}/\mathbb{Z}$. It is clearly infinite since $\frac{1}{p^i} + \mathbb{Z}$ are distinct elements.
   (3) For any $x = \frac{a}{p^i} + \mathbb{Z} \in \mathbb{Z}(p^\infty)$, $p^i x = 0$ so $|x| \mid p^i$ is a power of p.
   (4) $H_n = \left\{\frac{a}{p^n} + \mathbb{Z} : a \in \mathbb{Z}\right\}$ is naturally isomorphic to $\{a + p^n\mathbb{Z} : a \in \mathbb{Z}\}$ which is $\mathbb{Z}/p^n\mathbb{Z}$.
   (5) Any element $\frac{a}{p^n} + \mathbb{Z}$ is contained in such a subset H_n.
   (6) For any $x = \frac{a}{p^k} + \mathbb{Z}$ and $n = p^l \cdot m$ with $\gcd(m, p) = \gcd(a, p) = 1$, suppose $t \equiv a^{-1} \pmod{p^{k+l}}$, then $y = \frac{t}{p^{k+l}} + \mathbb{Z}$ satisfy $x = ny$.

1.4.5

Prove the fourth isomorphism theorem (on your own). Let G be a group and let N be a normal subgroup of G; write $\pi : G \to \overline{G} = G/N$ for the projection. Show that there is a bijection

$$\begin{array}{ccc} \{\text{subgroups } A \text{ of } G \text{ containing } N\} & \longleftrightarrow & \{\text{subgroups of } \overline{G}\} \\ A & \longmapsto & \overline{A} := A/N \\ \pi^{-1}(C) & \longleftarrow & C. \end{array}$$

In addition, prove that, for all $A, B \leq G$ with $N \leq A$ and $N \leq B$:

1. $A \leq B$ if and only if $\overline{A} \leq \overline{B}$.
2. If $A \leq B$, then $[B : A] = [\overline{B} : \overline{A}]$.
3. $\overline{\langle A, B \rangle} = \langle \overline{A}, \overline{B} \rangle$.
4. $\overline{A \cap B} = \overline{A} \cap \overline{B}$.
5. $A \trianglelefteq G$ if and only if $\overline{A} \trianglelefteq \overline{G}$.
   Proof: (0) This is clearly a well-defined map, and $\pi^{-1}(A/N) = A, \pi^{-1}(C)/N = C$, so it is a bijection.
   (1) $\overline{A} < \overline{B}$ iff $A/N < B/N$ iff $A < B$.

(2) If $B = \bigsqcup g_i A$ then $\overline{B} = \bigsqcup g_i A/N = \bigsqcup g_i \overline{A}$ since $g_i \overline{A}$ are disjoint so $[B : A] = [\overline{B} : \overline{A}]$.
(3)(4) Just verify at set-level.
(5) $\overline{A} \triangleleft \overline{G}$ iff $\forall a + N \in \overline{A}, g + N \in \overline{G}$ we have $gag^{-1} + N \in \overline{A}$, i.e. $gag^{-1} \in A \forall a \in A, g \in G$, which is just $A \triangleleft G$.

1.4.6

Show that

$$N = \{(1), (12)(34), (13)(24), (14)(23)\}$$

is a normal subgroup of S_4 contained in A_4. Give explicit isomorphisms

$$S_4/N \cong S_3 \qquad \text{and} \qquad A_4/N \cong \mathbb{Z}/3\mathbb{Z}.$$

Proof: Let $a = (12)(34), b = (13)(24), c = (14)(23)$, and consider the action of S_4 on $\{a, b, c\}$: $\sigma \cdot (ij)(kl) = (\sigma(i)\sigma(j))(\sigma(k)\sigma(l))$. This defined an epimorphism $S_4 \to S_3$ with kernel N, so $N \triangleleft S_4$ and $S_4/N \cong S_3$. Analogously we have an epimorphism $A_4 \to A_3$ with kernel N, so $A_4/N \cong \mathbb{Z}/3\mathbb{Z}$.

1.4.7

The set of $n \times n$ matrices can be identified with the space $\mathbb{R}^{n \times n}$. Let G be a subgroup of $\mathrm{GL}_n(\mathbb{R})$. Prove the following:

1. If $A, B, C, D \in G$, and if there are paths in G from A to B and from C to D, then there is a path from AC to BD. Here, a path from A to B means a continuous function $f : [0, 1] \longrightarrow G$ such that $f(0) = A$ and $f(1) = B$.
2. The set of matrices that admits a path to the identity I forms a normal subgroup of G (called the connected component of G), often denoted by G°.
3. Show that $\mathrm{SL}_n(\mathbb{R})$ is generated by elementary matrices of the form $I + aE_{ij}$ for $a \in \mathbb{R}$, $i \neq j$, and E_{ij} the $n \times n$ matrix with all zero entries except at the (i, j)-entry, where it is 1.
4. Use the above to deduce that $\mathrm{SL}_n(\mathbb{R})$ is path-connected; that is, all elements are linked to I_n by paths.
5. Show that the subgroup of connected component $\mathrm{GL}_n(\mathbb{R})^\circ$ in $\mathrm{GL}_n(\mathbb{R})$ has index 2.
   Proof: (1) Take paths $f, g : [0, 1] \to G$ connecting A, B and C, D, then $f \cdot g : [0, 1] \to G$ is a path connecting AC, BD.
   (2) If $f, g : [0, 1] \to G$ connects I, A and I, B, then (since $A \mapsto A^{-1}$ is continuous) $f \cdot g^{-1}$ is a path connecting I, AB^{-1}, so $G^\circ < G$.
   If $f : [0, 1] \to G$ connects I, A, and $B \in G$, then $GfG^{-1} : t \mapsto Gf(t)G^{-1}$ is a path connecting I, BAB^{-1} so $G^\circ \triangleleft G$.
   (3) By Gaussian elimination suppose $A \in \mathrm{SL}_n(\mathbb{R})$ is diagonal: $A = \mathrm{diag}\{d_1, \cdots, d_n\}$. Let $\varphi_{ij}(t) = (I + tE_{ij})(I - t^{-1}E_{ji})(I + tE_{ij})$ then $\varphi_{ij}(t)\varphi_{ij}(-1)$ gives the matrix $\mathrm{diag}\{t, t^{-1}, 1, \cdots, 1\}$. Clearly A can be decomposed into the product of these diagonal matrices.
   (4) $I + aE_{ij}$ is connected to I via the path $t \mapsto I + taE_{ij}$, so each generator $I + aE_{ij} \in G^\circ$ which implies $G = G^\circ$ is path-connected.

(5) If $M \in \mathrm{GL}_n(\mathbb{R})$ has $a = \det M > 0$, then $t \mapsto a^{t/n} M$ connected M to an element of $\mathrm{SL}_n(\mathbb{R})$, so the subgroup $\mathrm{GL}_n^+(\mathbb{R})$ of index 2 is contained in $\mathrm{GL}_n(\mathbb{R})^\circ$. Note that $\det(\mathrm{GL}_n(\mathbb{R})) = \mathbb{R} \setminus \{0\}$ is disconnected, so $\mathrm{GL}_n(\mathbb{R})^\circ$ is a proper subgroup. Therefore $\mathrm{GL}_n(\mathbb{R})^\circ = \mathrm{GL}_n^+(\mathbb{R})$ is a subgroup of index 2.

1.4.8

Show that $\mathrm{SL}_2(\mathbb{Z})$ is generated by the two matrices

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \text{and} \qquad S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Proof: Suppose $ad - bc = 1$, then $\gcd(a, c) = 1$ so we can use the Euclidean algorithm, by repeatedly multiplying T, T^{-1}, S, S^{-1} to turn (a, c) into $(1, 0)$. Now $d = 1$ so the resulting matrix is just T^b.

1.4.9

Let S and T be non-abelian finite simple groups, and write $G = S \times T$.

1. Show that the total number of normal subgroups of G is four.
2. If S and T are isomorphic, show that G has a maximal proper subgroup not containing either direct factor (meaning $\{1\} \times T$ or $S \times \{1\}$). A maximal proper subgroup of G is a subgroup H of G such that $H \neq G$ and the only subgroup of G strictly containing H is G itself. The existence of maximal proper subgroups is clear in this context because G is finite; you need to show that there is one such maximal proper subgroup that does not contain either direct factor.
3. If G has a maximal proper subgroup that contains neither of the direct factors of G, show that S and T are isomorphic.
   Proof: (1) Clearly $\langle e \rangle, S, T, G$ are four normal subgroups. If $H \triangleleft G$ is another normal subgroup, $H \cap S = 1$ or S and $H \cap T = 1$ or T. If $S \times 1 < H$, then H must be in the form $S \times K$ a contradiction, hence $H \cap S = 1$ and likewise $H \cap T = 1$. However, $[S, H] \subset S \cap H = 1$ so H commutes with S, and likewise H commutes with T, so every element in G commutes with H and $H < C(G)$. However S, T are non-abelian groups, so $S \times T$ has trivial center, a contradiction.
   (2) Consider the diagonal $\Delta = \{(s, s) : s \in S\}$ (assume $S = T$), which is clearly a proper subgroup. If there is a larger subgroup $\Delta < H < G$, suppose $(s, t) \in H \setminus \Delta$ then $(1, ts^{-1}) \in H$ with $a = ts^{-1} \neq 1$. Hence conjugating by elements of Δ, we know that $1 \times T < H$, a contradiction.
   Therefore the diagonal is a maximal proper subgroup not containing the factors.
   (3) Suppose $H < G$ is a proper maximal subgroup not containing S, T, then HS and HT are larger subgroups, so $HS = HT = G$ and hence $\pi_S(H) = S$, $\pi_T(H) = T$. Therefore $H \cap S = H \cap T = 1$ as they are normal subgroups of S, so $\pi_S|_H$ and $\pi_T|_H$ gives isomorphisms $S \cong H \cong T$.

1.4.10

Let G be a group and let $g \in G$ be an element of finite order n. Suppose that n is the product of two positive integers r and s which are coprime to each other.

1. Show that there is a pair (g_1, g_2) of elements of G such that $g_1^r = g_2^s = 1$ and $g_1 g_2 = g_2 g_1 = g$.

2. Show that such a pair is unique.
   Proof: (1) Take $u, v \in \mathbb{Z}$ such that $ur + vs = 1$, then $g_1 = g^{vs}$ and $g_2 = g^{ur}$ satisfy $g_1^r = g^{vsr} = 1$ and $g_2^s = 1$, and also $g_1 g_2 = g_2 g_1 = g^{ur+vs} = g$.
   (2) Note that g_1, g_2 commute, and g_2^{-1}, g commute, so $g_1 = g_1^{ur+vs} = g_1^{vs} = (g g_2^{-1})^{vs} = g^{vs}$ and likewise $g_2 = g^{ur}$.